WORKED SOLUTION

STEP I 1993, Question 4

Pure • Integration by Substitution

STEP I 1993, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q4

The key idea

The substitution $t = \pi - x$ reflects the interval $[0, \pi]$ onto itself, and it leaves $\sin x$ unchanged, because $\sin(\pi - t) = \sin t$. So the awkward factor x turns into $\pi - t$, and the integral we want appears again on the right-hand side with a minus sign. That lets us solve for it. For the integrals that follow, the main job is to write $1 + \cos^2 x$ in terms of $\sin x$. Over $[\pi, 2\pi]$, a shift by π brings us back to $[0, \pi]$.

Solution

The general result

Let $I = \displaystyle\int_0^{\pi} x\,\mathrm{f}(\sin x)\,\mathrm{d}x$ and put $t = \pi - x$. Then $\mathrm{d}x = -\mathrm{d}t$. When $x = 0$, $t = \pi$, and when $x = \pi$, $t = 0$. Also $\sin x = \sin(\pi - t) = \sin t$. Hence

$$\begin{aligned} I &= \int_{\pi}^{0} (\pi - t)\,\mathrm{f}(\sin t)\,(-\mathrm{d}t) = \int_0^{\pi} (\pi - t)\,\mathrm{f}(\sin t)\,\mathrm{d}t \\ &= \pi \int_0^{\pi} \mathrm{f}(\sin t)\,\mathrm{d}t - \int_0^{\pi} t\,\mathrm{f}(\sin t)\,\mathrm{d}t. \end{aligned}$$

The last integral is I again, since the name of the variable of integration doesn't matter. So $2I = \pi \int_0^{\pi} \mathrm{f}(\sin t)\,\mathrm{d}t$, which gives

$$\int_0^{\pi} x\,\mathrm{f}(\sin x)\,\mathrm{d}x = \frac{\pi}{2} \int_0^{\pi} \mathrm{f}(\sin x)\,\mathrm{d}x,$$

as required. (We need the integrals to exist, for example with f continuous on $[-1, 1]$.)

The integral over $[0, \pi]$

Ask yourself The integrand involves $\cos^2 x$, not just $\sin x$. Is it really of the form $x\,\mathrm{f}(\sin x)$?

Yes. Since $\cos^2 x = 1 - \sin^2 x$, we have $1 + \cos^2 x = 2 - \sin^2 x$, so

$$\frac{x \sin x}{1 + \cos^2 x} = x\,\mathrm{f}(\sin x) \qquad \text{with} \qquad \mathrm{f}(s) = \frac{s}{2 - s^2},$$

and f is continuous on $[-1, 1]$. By the general result,

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x}\,\mathrm{d}x = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x}\,\mathrm{d}x.$$

The remaining integral is easy because $\sin x$ is (minus) the derivative of $\cos x$. Substituting $u = \cos x$, with $\mathrm{d}u = -\sin x\,\mathrm{d}x$,

$$\int_0^{\pi} \frac{\sin x}{1 + \cos^2 x}\,\mathrm{d}x = \int_1^{-1} \frac{-\mathrm{d}u}{1 + u^2} = \int_{-1}^{1} \frac{\mathrm{d}u}{1 + u^2} = \Big[\arctan u\Big]_{-1}^{1} = \frac{\pi}{4} - \Big(-\frac{\pi}{4}\Big) = \frac{\pi}{2}.$$

Therefore

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x}\,\mathrm{d}x = \frac{\pi}{2} \cdot \frac{\pi}{2} = \frac{\pi^2}{4}.$$

The integral over $[0, 2\pi]$

The general result only covers $[0, \pi]$, so we split the range:

$$\int_0^{2\pi} = \int_0^{\pi} + \int_{\pi}^{2\pi}.$$

Ask yourself How can we bring $\int_{\pi}^{2\pi}$ back to an integral over $[0, \pi]$?

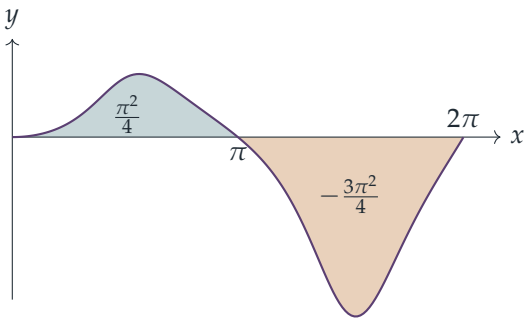
Shift by π: put $x = u + \pi$, so u runs from 0 to π. Then $\sin x = -\sin u$ and $\cos x = -\cos u$, so $\cos^2 x = \cos^2 u$. Hence

$$\begin{aligned}
\int_{\pi}^{2\pi} \frac{x \sin x}{1 + \cos^2 x}\,\mathrm{d}x &= \int_0^{\pi} \frac{(u + \pi)(-\sin u)}{1 + \cos^2 u}\,\mathrm{d}u \\
&= -\int_0^{\pi} \frac{u \sin u}{1 + \cos^2 u}\,\mathrm{d}u - \pi \int_0^{\pi} \frac{\sin u}{1 + \cos^2 u}\,\mathrm{d}u \\
&= -\frac{\pi^2}{4} - \pi \cdot \frac{\pi}{2} = -\frac{3\pi^2}{4},
\end{aligned}$$

using both results from the previous part. Adding the two pieces,

$$\int_0^{2\pi} \frac{x \sin x}{1 + \cos^2 x}\,\mathrm{d}x = \frac{\pi^2}{4} - \frac{3\pi^2}{4} = -\frac{\pi^2}{2}.$$

The graph shows why the answer is negative. On $[\pi, 2\pi]$ the factor $\sin x$ is negative, and there the factor x is larger than it is on $[0, \pi]$. So the area below the axis outweighs the area above it.



Checks

- Take $f \equiv 1$: the general result says $\int_0^{\pi} x\,dx = \frac{\pi}{2} \cdot \pi = \frac{\pi^2}{2}$, which is correct.
- The integrand is positive on $(0, \pi)$, so the first answer must be positive, and $\pi^2/4 \approx 2.467$ is. Simpson's rule gives 2.4674011 for the first integral and -4.9348022 for the second, matching $\pi^2/4$ and $-\pi^2/2$.
- The general identity also checks numerically for several random choices of f.

Common slips

- Applying the general result directly on $[0, 2\pi]$. It relies on the reflection $x \mapsto \pi - x$ mapping the interval to itself, and on $[0, 2\pi]$ it gives the wrong answer $\frac{\pi}{2} \cdot 0 = 0$ (the correct reflection there is $x \mapsto 2\pi - x$, under which $\sin x$ changes sign).
- Losing the minus sign from $dx = -dt$, or forgetting to swap the limits back.
- Writing $\arctan(-1) = 3\pi/4$: the arctangent takes values in $(-\pi/2, \pi/2)$, so $\arctan(-1) = -\pi/4$.

Beyond STEP

The trick here is to exploit a symmetry of the domain that the integrand nearly respects: replacing x by $\pi - x$ and averaging the two versions of the integral. The same idea gives $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x}\,dx = \frac{\pi}{4}$ without any integration. At university, symmetry arguments like this turn up throughout analysis and in Fourier series, where the parity of a function decides which terms survive. Siklos, *Advanced Problems in Mathematics*, has more STEP-style practice with substitutions of this kind, and Abbott, *Understanding Analysis*, makes the change-of-variables rule rigorous.

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